

## Amendment Verification Report

### VPF & Dynamic Agent Synthesis Architectural Consistency Amendment — IN DEVELOPMENT / FOR HUMAN REVIEW

#### 1. Chapter Structure Preserved

All 32 chapters are present, in the original order, with no renumbering. Programmatic validation (chapter-marker scan) confirms chapters 1-32 found exactly once each (chapter\_check = OK).

#### 2. Affected Chapters Only

A structural diff between v0.4 (1,123 non-TOC-cache body paragraphs) and v0.5 (1,227) found exactly 30 diff blocks. Every block falls within the front matter or one of the seven authorized chapters (1, 19, 20, 24, 28, 31, 32). The apparent large paragraph-count swing in a naive raw diff (1,457 old vs 1,227 new) is fully explained by Word's cached TOC-field text present only in the v0.4 PRODUCTION\_READY edition (it had previously been opened and saved in Word); once TOC-cache paragraphs are excluded from both editions, the comparison is clean and content-only.

#### 3. No Unrelated Architecture Changes

Chapters 2-18, 21-23, 25-27, 29-30 are byte-for-byte unchanged. Model Gateway (Ch.5), Tool Gateway (referenced throughout Ch.19-25), domain service architecture, security/privacy architecture (Ch.26), and Memory/Context architecture (Ch.20, cross-referencing NCIE-006) were not modified beyond the specific cross-references listed in the Change Register.

#### 4. Agent Factory Incorporated

Ch.24 §24.4A.1 recognizes the Agent Factory as a logical capability for governed Agent Definition construction, not a required physical microservice, and fixes the Codex/engineering boundary (Codex builds and maintains the Factory and new engineering primitives; ARGUS uses the Factory).

#### 5. VPF Gate A / Gate B Incorporated

Ch.24 §24.4A.5 places Gate A before novel Ephemeral Agent activation (envelope validation only); §24.4A.6 places Gate B after substantive AI generation and before validated release, applying to ARGUS direct model use, Registered Agents, Ephemeral Agents and multi-agent synthesis alike. The two gates are kept explicitly distinct, per directive §7.

#### 6. Ephemeral Agent Runtime Incorporated

Ch.24 §24.4A.3 defines the logical execution boundary enforcing Task Scope, Current Authorization, Permitted Models/Tools/Data/Context, Resource Limits, Network/Egress, Lifetime and Termination.

## **7. Control Plane / Execution Plane Incorporated**

Ch.24 §24.4A.8 distinguishes the Orchestration Control Plane (discovery, synthesis request, lifecycle, registry state, eligibility, policy, risk, VPF requirements, suspension, routing) from the Agent Execution Plane (executes the bounded workload only, with no administrative authority over any Registry, VPF Policy or Agent Factory governance), illustrated with a relationship diagram. No physical deployment is prescribed beyond NCIE-004.

## **8. Degraded Modes Incorporated**

Ch.28 §28.3 adds four new Dependency Degradation Matrix rows (Agent Factory, Ephemeral Agent Runtime, VPF, Registry unavailable), each naming what remains available versus what degrades explicitly. VPF FAILURE SHALL NOT FAIL OPEN is stated in Ch.24 §24.4A.4 and cross-referenced from Ch.28. Model/provider fallback doctrine (Ch.28 §28.3 existing AI/Model Gateway row; Ch.24 §24.9) is unchanged.

## **9. Codex Boundary Incorporated**

Ch.24 §24.4A.1 and Ch.32 §32.4A both state: Codex/engineering builds and maintains the Agent Factory and new engineering primitives; ARGUS uses the Factory to synthesize bounded agents from approved primitives without bespoke Codex development for each individual agent; Codex re-enters only for a capability exceeding the approved envelope (new Tool, Connector, Runtime Capability, Security Primitive, Persistent Service, Infrastructure or material Production Engineering Change).

## **10. Human-Primary Preserved**

Ch.1 §1.3.1 (Human-Primary Authority) is unchanged. Ch.31 §31.3B adds the mandatory adversarial acceptance scenario confirming that Gate A pass, sandbox pass, eligible model/tool use, Gate B PASS and multi-agent agreement do not, individually or together, create institutional execution authority absent current human/institutional authorization.

## **11. Model Gateway / Tool Gateway Preserved**

Ch.24 §24.4A.10 restates AGENT → MODEL GATEWAY → ELIGIBLE MODEL (never an arbitrary provider) and AGENT → TOOL GATEWAY → AUTHORIZED TOOL (never direct consequential system access), and states explicitly that Dynamic Agent Synthesis creates no bypass path for either gateway.

## **12. Memory Boundary Preserved**

Ch.24 §24.4A.10 states agents obtain governed context exclusively through the existing NCIE Memory/Context architecture (Ch.20); no unrestricted Institutional Memory access, independent permanent agent memory, or agent ownership of NCIE Memory is depicted. AGENT RUNTIME STATE ≠ INSTITUTIONAL MEMORY; detailed Memory semantics remain in NCIE-006.

## **13. Traceability Complete**

Every amended chapter's existing Traceability line was extended (not replaced) to cite NCIE-001 v1.3 and/or NCIE-007 v1.1 alongside its original NCIE-001 v1.2 reference. No new or competing traceability scheme was introduced.

## **14. Diagram Verification**

27 diagrams are present after amendment (up from 23 in v0.4), reflecting the 4 new diagrams added in Ch.24 §24.4A.8-§24.4A.9 (1 relationship diagram, 3 flow diagrams). Existing diagram captions in Ch.28 and Ch.32 were renumbered automatically by the shared figure counter (Figure 22→26, Figure 23→27) as a direct, expected consequence of the new figures inserted earlier in the document — not a content change. No diagram implies agent → arbitrary model provider, agent → direct consequential external system, agent → unrestricted Memory, agent → Agent Factory recursive creation, agent → self-registration, or VPF → institutional Decision authority.

## **15. Stale-Contradiction Search Completed**

A full-document keyword search (directive §25) found 22 matches, all from this amendment's own new content correctly stating the approved architecture. No pre-existing contradicting statement was found. No additional surgical amendment was required.

## **16. No Unrelated Amendment Creep**

No architecture wording, domain design, data architecture, security architecture, Memory architecture, voice architecture, revenue architecture, Anti-Fraud architecture, REWS architecture or formatting was modified outside the seven authorized chapters and front matter.

## **17. Technology Neutrality Preserved**

Ch.24 §24.4A's introductory paragraph explicitly states that no specific Agent Framework, container runtime, model provider, sandbox product, workflow engine, VPF software or registry product is named as architectural truth; NCIE-004 governs technology selection.

## **18. Human Review Items**

No new Human Review item was manufactured. No Agent Approval Board, VPF Committee, specific security owner, specific Agent promotion role or operational SLA value was invented; these remain open institutional decisions tracked in NCIE-007 v1.1 Chapter 35 and are only referenced, not duplicated or resolved, here.

## **19. Document-Control Correction**

A subsequent human review identified two stale document-control statements, both pre-dating this amendment: (1) the cover metadata line 'Traces to NCIE-001 Master Product Requirements Document, Version 1.2' had not been bumped to v1.3 alongside the governance-page 'Authoritative source' row, which already correctly cited v1.3; and (2) the governance-page 'Supersedes' field still read 'NCIE-002 Skeletal Architecture Review Edition v0.4' (accurate for v0.4's own governance page, describing what v0.4 itself superseded) while a separately added 'Prior baseline' row correctly

identified v0.5's actual prior edition as 'NCIE-002 v0.4 Production Ready' — the two fields were inconsistent.

Correction applied: the cover line now reads 'Traces to NCIE-001 Master Product Requirements Document, Version 1.3'. The Supersedes field now reads 'NCIE-002 v0.4 Production Ready (17 August 2026), which itself superseded the Skeletal Architecture Review Edition v0.4 (chapter structure preserved throughout)' — correctly identifying what v0.5 supersedes while preserving the intentionally historical mention of the earlier skeletal edition. The per-chapter 'status' lines that cite specific NCIE-001 v1.2 chapter names (e.g. 'Traces to NCIE-001 v1.2 Product Vision...') were correctly left untouched — these are accurate historical citations of the v1.2 chapters each NCIE-002 chapter was originally built against, not the current-document-authority statement the review flagged. A rebuild produced identical paragraph and table counts (1,227 paragraphs, 66 tables) to the pre-correction v0.5 edition, and both corrected fields were confirmed present at their expected locations — confirming this was a document-control-only fix with no substantive drift.

## **20. Disposition**

NCIE-002 — System Architecture & Technical Design v0.5 — VPF & Dynamic Agent Synthesis Architectural Consistency Amendment is READY FOR HUMAN REVIEW. The v0.4 approved baseline remains historically preserved and in force except for the specific amendments listed in the accompanying Change Register. This document is NOT marked Approved.